

GASOLINE DEMAND IN THE OECD

An Application of Pooling and Testing Procedures

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This study utilizes a pooled inter-country data set, finding the long-run price-elasticity falls in the range -0.55 to -0.9 , depending on the choice of pooled estimators. The estimators included the OLS, within-, and between-country estimators, plus five feasible GLS estimators. Even allowing for a ten-year distributed lag on price to reflect changes in auto-efficiency characteristics, the within-country estimator yields appreciably more inelastic estimates than did the OLS estimator, which was heavily influenced by the between- or inter-country variation. This difference raises intriguing questions for future research.

1. Introduction

Unfortunately, the theoretical econometrics literature fails to provide any clearcut directives to applied researchers in the choice of whether to pool cross-section/time-series data and the appropriate technique. The theoretical justification given for pooling rests on the finding that the estimators from a pooled model will in general be more efficient than those based on individual time series.¹ Questions of bias do not arise, as both individual time-series and pooled cross-section/time-series models yield unbiased estimators. Given the decision to pool, the choice of estimation techniques again rests on relative efficiency proofs. Such findings appear to have had little effect on research in the area of energy demand in view of the wide diversity of pure time-series, cross-sectional and pooled inter- and intra-country studies.² Moreover, the findings from these alternative data sets have seldom led to unanimity.³

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¹See Balestra and Nerlove (1966).

²For a review of electricity demand studies, see Taylor (1975). For other types of energy, see Pindyck (1980).

³For an example of conflicting estimates of energy-capital substitution response, see Berndt and Wood (1975) and Griffin and Gregory (1976).

The purpose of this paper is to provide applied evidence on the questions of (a) whether pooling is beneficial, (b) the selection of the appropriate pooled estimation technique, and (c) the interpretation of various tests applied to pooled models. The advantages of posing these questions in the framework of an applied study are that economic theory and knowledge of institutional differences provide important checks against the econometric results. We consider these broader questions within the context of analyzing gasoline demand for a sample of eighteen OECD countries with annual observations for the period 1960 to 1978. The magnitude of the price elasticity of gasoline is an important unresolved issue in itself. To date, most econometric studies report very low price elasticities ranging from -0.04 to -0.24 . For example Houthakker, Verleger and Sheehan (1974), using quarterly time-series data for a cross-section of U.S. states, obtained a long-run elasticity of -0.24 .⁴ There is not unanimity, however, as Sweeney (1978), using a model which explicitly introduced the efficiency characteristics of the auto fleet, found a long-run price elasticity of gasoline of -0.73 . An important research question is whether or not pooled inter-country data tend to support the Sweeney findings.

In section 2, we outline a simple theoretical model which forms the basis for econometric estimation. Specifically, this model describes how gasoline price and income affect the utilization and the efficiency of the auto fleet. Both static and dynamic models are postulated. Section 3 outlines the following data sets for which price and income elasticities are obtained: individual-country time series, within-country pooled model, between-country model, OLS pooled data, and five alternative methods of obtaining feasible generalized least-squares estimators (GLS). The two-stage GLS techniques for estimating error components models include techniques by Amemiya (1971), Wallace and Hussain (1969), Swamy and Arora (1972), Nerlove (1971) and Rao's (1972) MINQUE estimator. Section 4 examines the results in a static formulation, first asking question (a) — whether pooled models offer different estimators than individual-country time series. The analysis then turns to question (b) and a comparison of alternative pooling procedures. Section 5 replicates a similar exercise except here we consider dynamic specifications. In section 6, we subject our models to a battery of tests applicable to pooled GLS models. Section 7 recapitulates the major findings.

2. Model specification

Following Sweeney (1978) and Griffin (1979), our approach to modeling gasoline consumption is to view consumption as composed of three separate determinants: the utilization of the typical auto, its gasoline efficiency, and

⁴Other supporting evidence in this elasticity range is found in Philips (1972), Chamberlain (1973), Mehta, Narasimham and Swamy (1978), and Wharton (n.d.).

the stock of cars on the road. These three factors form the following consumption identity:

$$\begin{aligned} \text{Gasoline consumption} &= \text{Miles driven per car} \times \text{Gasoline consumption per mile} \times \text{Cars} \\ &\left(\text{Utilization} \times \frac{1}{\text{Efficiency}} \times \text{Stock of cars} \right). \end{aligned} \quad (1)$$

The advantage to analyzing gasoline consumption in these three components is that short- and long-run effects can be separated. For example, changes in auto efficiency require a long enough period to turn over the stock of cars. On the other hand, changes in the utilization of cars may be achieved in the short run with the existing auto stock.

International data limitations prevent this approach from being fully implemented, however. The lack of data on miles driven and gasoline consumption per mile requires that the utilization and efficiency factors be combined, leaving gasoline consumption per auto (GAS/CAR) to be explained by variables reflecting utilization (U) and efficiency (E),

$$GAS/CAR = U/E. \quad (2)$$

The empirical determination of eq. (2) is based on variables that reflect the utilization of autos, such as per capita income (Y/N) and gasoline price (P_{MG}/P_{GDP}). In addition to these economic variables, the rising stock of cars per capita (CAR/N) is likely to lead to reduced auto utilization. For example, the two-car family does not drive twice the miles of a one-car family. Since gasoline efficiency is held constant, we enter efficiency (E) explicitly. Eq. (2) can be written in estimation form as follows:

$$\begin{aligned} \ln(GAS/CAR) &= \alpha_0 + \alpha_1 \ln(Y/N) + \alpha_2 \ln(P_{MG}/P_{GDP}) \\ &+ \alpha_3 \ln(CAR/N) + \alpha_4 \ln E. \end{aligned} \quad (3)$$

In turn, the gasoline efficiency (E) of the existing auto fleet follows a putty-clay model and is largely determined by the technical characteristics of the autos at the time they enter the auto stock. In any given model year, we hypothesize that per capita income (Y/N) and the price of gasoline (P_{MG}/P_{GDP}) are likely to determine the gasoline efficiency of autos entering the stock. Since the auto stock has a gestation period of eight to ten years, gasoline efficiency (E) will depend on distributed lags (L), on per capita income (Y/N), and gasoline price (P_{MG}/P_{GDP}),

$$\ln E = \delta + \gamma(L) \ln(Y/N) + \beta(L) \ln(P_{MG}/P_{GDP}). \quad (4)$$

Since gasoline efficiency is not directly measurable, eq. (4) is substituted into (3) and rewritten as follows:

$$\ln(GAS/CAR) = \delta^* + \gamma^*(L) \ln(Y/N) + \beta^*(L) \ln(P_{MG}/P_{GDP}) + \phi^* \ln(CAR/N), \quad (5)$$

where δ^* , γ^* , β^* and ϕ^* are appropriately redefined. It should be noted that β_0^* reflects a short-run effect working primarily through adjustment of auto utilization, while lagged values of β^* reflect the previous adaptations of the auto fleet to changes in gasoline prices.

The principal empirical focus is on the price elasticity of gasoline demand and the adaptation process from the short to the long run. It should be noted that since our model treats the stock of cars as exogenous, our gasoline price elasticities reflect only price effects working through utilization and efficiency adjustments. To the extent that gasoline prices affect the number of autos in the stock, our estimates will understate the price elasticities of gasoline demand. As Sweeney (1978) demonstrates, stock effects are likely to be quite small compared to utilization and efficiency adaptations.

Our data set consists of a pooled sample for eighteen OECD countries covering the period 1960 to 1978. A major advantage to this sample is the wide variation in per capita incomes, relative gasoline prices, and cars per capita both over time and across countries. The sample includes five data points for each country following the 1973 Arab Oil Embargo and the price increases it triggered. Across countries, there are wide differences in relative gasoline prices owing to the substantial tax rate differences. A more-detailed description of the data, sources, and necessary adjustments for purchasing power parities is given in the appendix.⁵

For purposes of estimation, we employ a static and two competing dynamic specifications for eq. (5). The following model is essentially a static version of eq. (5):

Model I

$$\ln(GAS/CAR) = \delta^* + \gamma^* \ln(Y/N) + \beta^* \ln(P_{MG}/P_{GDP}) + \phi^* \ln(CAR/N). \quad (6)$$

⁵One major data complexity encountered in the estimation of eq. (5) arises from the fact that the motor-vehicle variable (*CAR*) consists of passenger cars. However, some motor gasoline is consumed by vehicles other than passenger cars — trucks and buses, for example. Since it is not possible to make explicit adjustments in the data, the variables 'truck per CAR' (*TRK/CAR*) and 'truck per capita' were introduced in an attempt to capture the effect of non-passenger car-gasoline-using vehicles. Also, to test for the effect of diesel fuels substitution against gasoline, the price of diesel fuel relative to motor gasoline was introduced. These variables proved to be statistically insignificant.

Even though the underlying model is clearly dynamic, the static specification is of interest because it indicates the degree to which the static coefficients approximate the short-run (e.g., $\beta^* = \beta_0^*$) or long-run (e.g., $\beta^* = \sum \beta_i^*$) responses.

Two dynamic specifications are contrasted. The first introduces the familiar Koyck distributed lag by the inclusion of the lagged dependent variable.⁶

Model II

$$\ln(GAS/CAR) = \delta^* + \gamma^* \ln(Y/N) + \beta^* \ln(P_{MG}/P_{GDP}) + \phi^* \ln(CAR/N) + \lambda \ln(GAS/CAR)_{-1}. \quad (7)$$

In this specification, the short-run price elasticity is given by β^* and the long-run elasticity is $\beta^*/(1-\lambda)$. The flow adjustment model is of particular interest because it follows the earlier work of Houthakker, Verleger and Sheehan (1974), and because of the estimation problem it poses in some pooled models.

The second specification features a polynomial distributed lag approach to the estimation of eq. (5) as follows:

Model III

$$\ln(GAS/CAR) = \delta^* + \sum_{i=0}^m \gamma_i^* \ln(Y/N)_{-i} + \sum_{i=0}^n \beta_i^* \ln(P_{MG}/P_{GDP})_{-i} + \phi^* \ln(CAR/N). \quad (8)$$

Second-degree polynomials constrained at the far end points are utilized to estimate the lag structures γ^* and β^* : the lengths of the lag distribution are selected on the basis of minimum standard error. The advantage of this approach vis-à-vis the Koyck formulation is the allowance for a more general lag structure and the use of only exogenous explanatory variables. The disadvantage is that even after being able to extend the income and price data back to 1955, we were still forced to work with a smaller sample than allowed by Model II.

3. Estimation techniques

Given annual time-series observations for each of the eighteen countries,

⁶Model II is included here for pedagogical purposes because of the frequent use of specifications involving lagged dependent variables. It is not intended to follow from eq. (6) since the cars-per-capita variable does not involve lags.

the choice of estimation techniques is by no means obvious. Our purpose here is to contrast the results from the most prominent alternatives. The first alternative is simply to rely on individual-country time-series estimates. In general notation, the individual time-series model is

$$Y_{it} = \alpha_i + \sum_k \beta_{i,k} X_{it,k} + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (9)$$

where i denotes the i th country, t denotes the t th time period.

Another alternative is to pool the data using an error-components model

$$Y_{it} = \alpha + \sum_k \beta_k X_{it,k} + u_{it} \quad \text{with} \quad u_{it} = \mu_i + v_{it}, \quad (10)$$

where μ_i is the i th individual, time-invariant, country effect and v_{it} is a remainder. The μ_i 's are assumed to be random with zero mean and variance σ_μ^2 and are independent of each other and the v_{it} 's.

Eight estimation methods are applied to (10). These include OLS, within- and between-country techniques, as well as five other two-stage GLS methods. The within regression is basically a least-squares regression with country dummies or alternatively performing OLS on the following equation:

$$Y_{it} - \bar{Y}_{i.} = \sum_k \beta_k (X_{it,k} - \bar{X}_{i.,k}) + (u_{it} - \bar{u}_{i.}), \quad (11)$$

where $\bar{Y}_{i.} = \sum_t Y_{it}/T$ is the analysis of variance notation for the i th country mean. The between-country regression is the regression of individual-country means, or equivalently performing OLS on the following equation:

$$Y_{i.}/\sqrt{T} = \alpha\sqrt{T} + \sum_k \beta_k X_{i.,k}/\sqrt{T} + u_{i.}/\sqrt{T}, \quad (12)$$

where $Y_{i.} = \sum_t Y_{it}$.

For this model, the GLS estimator of the regression coefficients can be obtained by performing OLS on the following transformed regression equation:⁷

$$(Y_{it} - \theta \bar{Y}_{i.}) = \alpha^* + \sum_k \beta_k (X_{it,k} - \theta \bar{X}_{i.,k}) + u_{it}^*, \quad (13)$$

where $\alpha^* = \alpha(1 - \theta)$ and u_{it}^* is the transformed disturbance term which now has a homoskedastic variance-covariance matrix $\sigma_v^2 I_{NT}$, where σ_v^2 is the

⁷See Hausman (1978) and Fuller and Battese (1974) for more details on how this transformation is obtained.

variance of v_{it} , and I_{NT} is an identity matrix of dimension NT , and $\theta = (1 - \sigma_v/\sigma_1)$ and $\sigma_1^2 = \sigma_v^2 + T\sigma_\mu^2$.

The five two-stage GLS procedures differ only with respect to their estimators of the variance components σ_μ^2 and σ_v^2 , and the corresponding θ . The Amemiya (1971) and Wallace and Hussain (1969) procedures obtain analysis of variance-type estimates of the variance components where the former method relies on within residuals and the latter relies on OLS residuals. The Swamy and Arora (1973) procedure estimates the variance components using mean square residuals of the within- and between-country regressions. The Nerlove (1971) procedure estimates σ_μ^2 as the sample variance of the country dummies and σ_v^2 as the residual sum of squares of the within regression divided by the number of observations. The MINQUE procedure constructs a minimum norm quadratic unbiased estimator of the variance components. For details on the construction of the MINQUE estimator, the reader is referred to Rao (1972).

For Models I and III, all estimation methods are unbiased. OLS is asymptotically inefficient while the within estimator is asymptotically efficient. Finite sample results⁸ and Monte-Carlo studies⁹ indicate that (i) as long as the variance components are not small and close to zero, there is gain in performing the two-stage GLS methods rather than OLS, the within, or the between estimators; (ii) according to the MSE criterion, there is little to choose among the two-stage GLS methods; (iii) better estimates of the variance components do not necessarily give better second-round estimates of the regression coefficients.

4. The static model results

In this section, we utilize Model I, a purely static formulation, to address question (a) pertaining to the gains from pooling. Table 1 presents the individual-country time-series estimates and table 2 reports the results using alternative pooling techniques. On balance, the signs of individual parameter estimates conform to *a priori* expectations. Income effects are positive while price effects are negative. Increasing cars per capita has tended to reduce auto utilization as expected. In only four instances are perverse signs observed, and these are not statistically significant. Nevertheless, the frequency of correctly signed, but not statistically significant coefficients provides little consolation to policy-makers. For example, in ten out of the eighteen countries, the income effect is not statistically significant and in seven countries the price of gasoline is not statistically significant.

The pooled results in table 2 provide some useful insights into the

⁸See Swamy (1971), Swamy and Arora (1972), Swamy and Mehta (1979), and Taylor (1980).

⁹See Nerlove (1971), Maddala (1971), Arora (1973), Maddala and Mount (1973), and Baltagi (1981).

Table 1
Model I parameter estimates from individual-country time series.^a

Country	$\ln(Y/N)$	$\ln(P_{MG}/P_{GDP})$	$\ln(CAR/N)$	Const.	$\bar{R}^2$	S.E.
Canada	0.39 (5.1)	-0.36 (4.1)	-0.44 (6.2)	3.13 (11.1)	0.791	0.012
U.S.	0.11 (0.8)	-0.28 (3.1)	-0.96 (0.8)	4.33 (15.4)	0.453	0.016
Japan	-0.05 (0.3)	-0.14 (2.5)	-0.56 (9.9)	1.22 (2.9)	0.999	0.026
Austria	0.76 (3.6)	-0.79 (5.3)	-0.52 (4.6)	3.73 (10.0)	0.680	0.039
Belgium	0.85 (5.0)	-0.04 (0.3)	-0.67 (7.2)	3.04 (6.7)	0.891	0.034
Denmark	0.93 (0.4)	-0.14 (0.9)	-0.52 (4.0)	0.24 (0.7)	0.964	0.030
France	1.14 (6.9)	-0.20 (2.1)	-0.86 (6.5)	3.17 (10.7)	0.735	0.026
Germany	0.40 (3.5)	-0.17 (2.6)	-0.22 (3.4)	4.26 (15.7)	0.492	0.017
Greece	0.59 (1.1)	-0.34 (2.3)	-0.47 (2.1)	3.69 (2.9)	0.914	0.075
Ireland	0.35 (1.6)	-0.099 (0.9)	-0.18 (2.0)	4.82 (5.0)	0.190	0.039
Italy	0.12 (0.7)	-0.37 (6.7)	-0.36 (6.2)	1.27 (2.1)	0.978	0.033
Netherlands	0.36 (1.3)	-0.40 (2.1)	-0.62 (6.9)	0.62 (0.7)	0.980	0.040
Norway	0.80 (3.6)	-0.23 (1.6)	-0.66 (6.0)	2.91 (7.2)	0.935	0.031
Spain	-0.83 (1.6)	-0.08 (0.5)	-0.10 (0.7)	-1.56 (1.0)	0.967	0.057
Sweden	-0.71 (1.8)	-0.62 (3.3)	0.04 (0.2)	-2.89 (1.5)	0.478	0.026
Switzerland	1.07 (5.2)	-0.40 (2.8)	-0.62 (8.8)	4.93 (7.2)	0.923	0.028
Turkey	0.32 (1.2)	-0.26 (1.8)	-0.60 (4.9)	0.48 (0.7)	0.929	0.087
U.K.	0.56 (2.6)	-0.06 (1.0)	-0.33 (4.1)	4.49 (6.5)	0.685	0.027

^aNumbers in parentheses are *t*-statistics.

situations in which pooling is beneficial. For the moment, we focus on the within-country estimation technique, as this method relies entirely on the intra-country variation characteristic of individual time series. Subsequently,

Table 2
Model I parameter estimates from pooled sample.^a

Estimation technique	$\ln(Y/N)$	$\ln(P_{MG}/P_{GDP})$	$\ln(CAR/N)$	Const.	$\bar{R}^2$	S.E.
OLS	0.89 (24.8)	-0.89 (29.4)	-0.76 (41.0)	2.39	0.854	0.210
Within	0.66 (9.3)	-0.32 (7.5)	-0.64 (22.1)	—	0.838	0.090
Between	0.97 (6.2)	-0.96 (7.3)	-0.80 (9.6)	11.08	0.854	0.857
<i>Two-stage GLS</i>						
Amemiya	0.60 (9.2)	-0.36 (8.8)	-0.62 (22.7)	2.19	0.834	0.092
Wallace and Hussain	0.54 (9.9)	-0.47 (12.0)	-0.61 (24.8)	1.91	0.823	0.099
Swamy and Arora	0.55 (9.4)	-0.42 (10.5)	-0.61 (23.8)	1.99	0.828	0.095
Nerlove	0.60 (9.2)	-0.36 (8.8)	-0.62 (22.7)	2.19	0.834	0.092
MINQUE	0.54 (9.7)	-0.45 (11.4)	-0.61 (24.4)	1.94	0.840	0.093

^aNumbers in parentheses are *t*-statistics.

we turn to a comparison of alternative pooling techniques. The pooled within results indicate an income elasticity of +0.66, a price elasticity of -0.32, and an elasticity of -0.64 with respect to cars per capita. These coefficients are all strongly significant, suggesting that the pooled model offers an impressive improvement in the efficiency of the parameter estimates.

What general observations do these results suggest? The gain in efficiency from pooling is large, in general, but particularly impressive for the income and price effects as distinct from the cars per capita variable, which with the exception of the U.S., Spain and Sweden, was statistically significant in the individual-country time-series results. The explanation for this phenomena offers important insights. The period 1960 to 1978 marked a period of sharply increasing cars per capita with the percentage increases ranging from 62.1% in the U.S. to 3711% in Japan. Given independent variation of this magnitude, it is not surprising that individual-country time-series tended to find significant effects of cars per capita. In contrast, both the magnitude and independence of the variation in the income and price variables was much more limited. The frequent insignificance of the income variable can be attributed both to the modest growth in real per capita incomes over the period 1960 to 1978 and to the extent it occurred, positively correlated with the growth in cars per capita. The Arab Oil Embargo of 1973 and

subsequent increases provided substantial real price increases, which explains the fact that the price effect is statistically significant in eleven of the eighteen countries. To illustrate the importance of the post-1973 price variation, individual-country time-series were calculated for the period 1960 to 1972, in which relative gasoline price variation was quite small. The price coefficient was significantly negative in only two of the eighteen countries. Results such as these offer a warning to time-series studies finding the absence of price and income effects. Low *t*-statistics are more an indicator of large confidence intervals than evidence of the absence of the economic effect. Conversely, it does not follow that a pooled sample guarantees efficient parameter estimates since even in pooled samples, the extent of independent variation in the explanatory variables may be limited. Fortunately, the issue of the magnitude and independence of the variation are issues which can be analyzed prior to sample selection.

Turning to the intriguing question of which pooled estimation technique is preferable, we note that table 2 offers a varied menu at least with respect to price-elasticity estimates. The price elasticity varies from -0.32 for the within estimator to -0.96 for the between-country estimator. Alternatively, the income-elasticity estimates vary from 0.54 with Wallace and Hussain to 0.97 for the between-country estimator. The elasticity with respect to cars per capita varies within a much more narrow range, 0.61 to 0.80 .

In interpreting the results from these various estimators, it is useful to first contrast the within and between estimators, since all of the other seven techniques are related to these two major sources of variation. The between-country variation indicates that the price elasticity is much greater (-0.96) in explaining inter-country differences in gasoline consumption per car. An estimator based purely on variation within countries indicates a much smaller response (-0.32). There is a natural tendency to affix a 'long-run' label to the between-country results and a 'short-run' label to the within-country results. The empirical results are entirely consistent with this interpretation, suggesting between-country variation reflects long-run responses, with each country having adapted its auto efficiency, utilization, and driving infrastructure to long-standing patterns of price and income. In contrast, a static model, applied to the pure time-series variation present in the within-country sample, seems likely to elicit primarily short-run responses. Since the OLS model weighs both within- and between-country observations equally even though the between-country error term has a much larger variance, the OLS estimators tend to conform to the between-country results. The various two-stage GLS techniques had parameter estimates closer to the within-country estimates because of the relatively high between variance compared to the within variance. It should be noted that the parameter estimates of the two-stage GLS techniques need not fall in the range between the OLS and within estimators as evidenced by the income

and cars per capita parameter estimates. In the single-explanatory-variable case, this frequently cited property holds as θ ranges between 0 (OLS) and 1 (within). However, with multiple explanatory variables, intercorrelations among the explanatory variables vitiate this property as indicated by table 2. Nevertheless, even though the parameter estimates need not fall between the OLS and within estimators, they obviously do provide reasonable *a priori* bases for tendencies around which the estimators will generally group.

In contrast to theoretical proofs that all five two-stage GLS estimators are asymptotically efficient, and to Monte-Carlo studies that report only minor small-sample differences among these estimators,¹⁰ the price elasticities range from -0.36 using Amemiya's procedure to -0.47 using Wallace and Hussain's procedure. These differences are even more striking when one notes that -0.36 does not even belong in the 95 percent confidence interval of the Wallace-Hussain estimate, nor that of MINQUE. Conversely, the 95 percent confidence interval of the Amemiya estimate of -0.36 does not include the Wallace-Hussain estimate of -0.47 , nor the MINQUE estimate of -0.45 . Only the Swamy-Arora estimate of -0.42 lies in the 95 percent confidence interval of any of the two-stage GLS estimators.¹¹ Thus in situations involving substantial differences in the OLS and within estimators, alternative GLS estimators can yield statistically different estimators. Good indicators of the range of variation are the Amemiya and Wallace-Hussain estimators. The Swamy-Arora estimator offers the 'most representative' estimates.

5. Dynamic models of gasoline demand

5.1. Model II results

The results of our dynamic demand model are particularly interesting in view of previous work indicating low long-run price elasticity and rapid adjustment to the long run. Houthakker, Verleger and Sheehan (1974) found a very rapid adjustment from the short run to the long run as the first-quarter price elasticity is -0.08 and the long-run elasticity is -0.24 . Similarly, Chamberlain (1973), using an aggregate time-series equation, found low price elasticities of -0.06 in the short run and -0.07 in the long run. Mehta, Narasimham and Swamy (1978) report a long-run price elasticity of -0.04 . Even though Ramsey, Rasche and Allen (1975) report a marginally significant long-run price elasticity of -0.8 , they find that adjustment occurs instantaneously.

¹⁰See the references cited in footnotes 8 and 9.

¹¹Based on W.E. Taylor (1980), the asymptotic variance of the feasible GLS estimators may be understated. Taylor's closest case is that where ($T=10$, $N-k=10$), the true variances in the latter case may be understated by at most 6 percent. Even after accounting for this worst possible understatement, the Amemiya estimator is still not within the 95 percent confidence interval of the Wallace and Hussain estimator.

These studies present a series of problems that may bias the price elasticities toward zero, emphasizing short-term rather than long-term developments. In order to avoid the collinearity and time-trend problems in aggregate time series, Houthakker and his colleagues have wisely chosen to work with pooled data, but the usual characteristics of pooled samples are not realized. The U.S. data over the period 1960 to 1971 present only a very small range of price variation between states and over time and permit little adjustment from the point of view of the technical characteristics of the vehicle itself. For many years, throughout the United States, the price of gasoline was relatively low and cars relatively cheap. The U.S. road system, spatial distribution of the population, and the automobile stock have adjusted to these characteristics, so that up until the mid 1970's, Americans have driven heavy 'gas-guzzling' cars over fairly large distances at relatively high average speeds. The American driver had little opportunity to drive other types of vehicles, since, until recently, the U.S. automobile industry did not offer small cars, and since many foreign cars were previously ill-adapted to U.S. driving conditions. For the U.S. sample, we cannot expect to see the potentially large adjustment both in the characteristics of vehicles and in driving conditions. Thus we would expect a tendency for the between variation to reflect long-run responses and the within data to reflect essentially a short-run response. The question then becomes: Will the addition of a dynamic specification be sufficient to enable the within-country data to capture long-run responses?

Table 3 reports the parameter estimates of Model II which utilizes the Koyck distributed lag. These results indicate much greater price elasticity than found in the above studies as the short-run price elasticity is of a similar magnitude as Houthakker, Verleger and Sheehan's long-run elasticity of -0.24 . The relatively high coefficient on the lagged dependent variable is indicative of a much longer adjustment period as well. For example, a coefficient of 0.8 indicates that only 54% of the adjustment to the long-run response occurs within 5 years. Long-lag structures are supportive of the view that adaptations in the gasoline efficiency of the auto fleet and driving conditions require long periods for adjustments. This result is entirely consistent with the vintage-capital approach of Sweeney (1978) who found elasticities of -0.22 in the first year, -0.50 after four years, and -0.73 after fourteen years.

Table 4 reports the long-run elasticity estimates implied by table 3. Compared to the static elasticity results of table 2, Model II yields substantially more elastic price responses. Such findings are to be expected in view of our between-country findings in table 2 which indicated that long-run inter-country differences in gasoline demand per car are substantially more price-elastic. The within technique yields a more elastic price elasticity, changing from -0.32 to -0.52 . Similarly, the two-stage GLS techniques

Table 3
Model II parameter estimates of pooled sample.^a

Estimation technique	$\ln(Y/N)$	$\ln(P_{MG}/P_{GDP})$	$\ln(CAR/N)$	$\ln(GAS/CAR)_{-1}$	Const.	$\bar{R}^2$	S.E.
OLS	0.07 (3.7)	-0.08 (4.7)	-0.04 (3.2)	0.93 (57.5)	0.254	0.987	0.062
Within	0.19 (4.1)	-0.16 (6.1)	-0.19 (7.2)	0.69 (23.6)	—	0.940	0.051
<i>Two-stage GLS</i>							
Amemiya	0.16 (5.0)	-0.17 (7.2)	-0.16 (7.6)	0.73 (26.9)	0.600	0.944	0.052
Wallace and Hussain	0.13 (5.6)	-0.14 (6.8)	-0.10 (5.9)	0.85 (39.9)	0.430	0.970	0.056
Swamy and Arora	0.13 (5.6)	-0.14 (6.8)	-0.11 (6.0)	0.84 (39.1)	0.440	0.969	0.056
Nerlove	0.16 (5.0)	-0.17 (7.2)	-0.16 (7.6)	0.73 (26.9)	0.601	0.944	0.052
MINQUE	0.14 (5.7)	-0.15 (7.0)	-0.11 (6.3)	0.83 (37.8)	0.457	0.967	0.055

^aNumbers in parentheses are *t*-statistics.

Table 4
Model II long-run elasticity estimators.

Estimation technique	Elasticity with respect to:		
	Y/N	P_{MG}/P_{GDP}	CAR/N
OLS	0.93	-1.10	-0.62
Within	0.63	-0.52	-0.60
<i>Two-stage GLS</i>			
Amemiya	0.61	-0.64	-0.60
Wallace and Hussain	0.84	-0.92	-0.67
Swamy and Arora	0.84	-0.91	-0.66
Nerlove	0.61	-0.64	-0.60
MINQUE	0.82	-0.89	-0.66

yield long-run price elasticities ranging from -0.64 to -0.92 as compared to the -0.36 to -0.47 range in table 2. With the exception of the within technique, Model II also yields more elastic income responses especially for the Wallace and Hussain, Swamy and Arora and MINQUE estimators which are now weighted closer to the OLS result, and therefore yielded substantially higher income and price elasticities.

In comparing differences among the two-stage GLS estimators, we note similar tendencies as those exhibited in table 2. The Amemiya and Nerlove estimators are quite similar and indicate a substantially lower long-run price elasticity. The MINQUE and Wallace and Hussain estimators likewise provide the most elastic response. One minor difference is that the Swamy and Arora estimator, which in table 2 fell in between the extremes, is here virtually identical to the MINQUE and Wallace and Hussain estimators.

Despite these intuitively more appealing elasticity results and a tendency for the within and GLS estimators to lie closer to the between estimators in table 2, Model II is subject to serious statistical difficulties. The OLS results are biased and inconsistent due to the combination of a lagged dependent variable and autocorrelated errors, which occur as a result of each country's errors falling above or below the regression line. Maddala (1971) has shown that the coefficient on the lagged dependent variables is biased towards unity while the remaining coefficients are biased toward zero. The within and GLS estimators are also biased, but they still have desirable asymptotic properties.

5.2. Model III results

Next we consider the sensitivity of the previous dynamic results to the introduction of polynomial distributed lags instead of the Koyck geometric lag structure. The long-run income and price coefficients are reported in table 5a with individual lag coefficients given in table 5b. Alternative runs with differing lag length, holding the degree of the polynomial at two and constraining the far end point to zero, indicated that on the basis of sum of squared errors, generally no lags on income and 9-year lags on price are preferred on theoretical and empirical grounds. As indicated in table 5b, the declining value of the lagged price terms is indicative of the declining frequency and utilization of older vintage autos.

Except for the within estimator, the long-run income and price elasticities obtained by Model III are smaller than those obtained by Model II. In view of the absence of lagged income effects, it is not surprising that the income elasticities in Model I tend to be quite close to those in Model III. For the within estimator, the long-run price elasticity (-0.55) was higher than that obtained by both Model I (-0.32) and Model II (-0.52), whereas the long-run income elasticity (0.54) was smaller than that obtained by both Model I (0.66) and Model II (0.63). For the two-stage GLS techniques, both price and income elasticities in Model III exhibited a narrower range than that obtained in Model I and Model II. Within these techniques, we again observe the tendency for the Amemiya and Nerlove estimators to offer the more price-inelastic estimate, while the MINQUE and Wallace and Hussain estimators yield the more elastic price response. Again, this is to be expected, since the Amemiya and Nerlove estimates of the variance components are

Table 5a
Model III parameter estimates of pooled sample.^a

Estimation technique	$\ln(Y/N)$	$\ln(P_{MG}/P_{GDP})^b$	$\ln(CAR/N)$	Const.	$\bar{R}^2$	S.E.
OLS	0.89 (24.3)	-0.90 (29.2)	0.77 (39.0)	2.34	0.86	0.191
Within	0.54 (6.8)	-0.55 (8.0)	-0.66 (20.3)	—	0.83	0.066
<i>Two-stage GLS</i>						
Amemiya	0.54 (8.6)	-0.59 (11.0)	-0.66 (22.7)	1.37	0.83	0.069
Wallace and Hussain	0.57 (10.3)	-0.63 (13.2)	-0.67 (24.4)	1.40	0.83	0.072
Swamy and Arora	0.55 (9.2)	-0.61 (11.7)	-0.66 (23.2)	1.37	0.83	0.069
Nerlove	0.54 (8.6)	-0.59 (11.0)	-0.66 (22.7)	1.37	0.83	0.069
MINQUE	0.55 (9.7)	-0.62 (12.4)	-0.67 (23.7)	1.38	0.83	0.070

^aNumbers in parentheses are *t*-statistics.

^bSee table 5b for individual lag coefficients.

based on the within residuals, whereas the MINQUE and Wallace and Hussain estimates of the variance components are based on the OLS residuals.

An intriguing question is: why are the price elasticities from Model III appreciably less elastic than the Koyck model (Model II)? Besides the estimation problems Model II imposes, another explanation is that the nine-year lag on price in Model III was simply not sufficient to reflect the full adaptation of auto efficiency and long-run utilization. The Koyck lag indicates ($\lambda=0.8$) that after nine years, only 70% of the long-run adjustment is realized. If indeed the differences explaining the between and within price-elasticity results of table 2 were that the within technique reflected essentially incomplete long-run adjustment, the introduction of nine-year distributed lags on price may be insufficient to reflect long-run adjustment. As discussed subsequently, this conjecture becomes important in testing for specification error.

6. Model tests

A series of tests were applied to our model. First, we tested the poolability of the data, given that the underlying model is a one-way error-component

Table 5b
Model III distributed lag coefficients.^a

Estimation technique	Coefficients on price									
	γ_0	γ_1	γ_2	γ_3	γ_4	γ_5	γ_6	γ_7	γ_8	γ_9
OLS	-0.160 (3.4)	-0.145 (5.0)	-0.130 (9.6)	-0.115 (29.1)	-0.099 (10.3)	-0.084 (5.4)	-0.068 (3.6)	-0.051 (2.8)	-0.034 (2.2)	-0.017 (1.9)
Within	-0.078 (3.9)	-0.077 (5.8)	-0.074 (8.0)	-0.070 (7.9)	-0.064 (6.3)	-0.057 (5.0)	-0.048 (4.1)	-0.038 (3.5)	-0.027 (3.1)	-0.014 (2.8)
Two-stage GLS										
Amemiya	-0.083 (4.0)	-0.082 (6.2)	-0.080 (9.6)	-0.076 (11.0)	-0.070 (8.6)	-0.062 (6.5)	-0.053 (5.2)	-0.042 (4.4)	-0.030 (3.8)	-0.016 (3.4)
Wallace and Hussain	-0.090 (4.2)	-0.088 (6.5)	-0.085 (10.6)	-0.081 (13.3)	-0.074 (10.1)	-0.066 (7.3)	-0.056 (5.7)	-0.045 (4.7)	-0.031 (4.1)	-0.017 (3.6)
Swamy and Arora	-0.085 (4.1)	-0.084 (6.3)	-0.082 (9.9)	-0.077 (11.8)	-0.071 (9.1)	-0.064 (6.8)	-0.054 (5.4)	-0.043 (4.5)	-0.030 (3.9)	-0.016 (3.5)
Nerlove	-0.083 (4.0)	-0.082 (6.2)	-0.080 (9.6)	-0.076 (11.0)	-0.070 (8.5)	-0.062 (6.5)	-0.053 (5.2)	-0.042 (4.4)	-0.030 (3.8)	-0.016 (3.4)
MINQUE	-0.087 (4.1)	-0.086 (6.4)	-0.083 (10.2)	-0.079 (12.5)	-0.072 (9.5)	-0.065 (7.0)	-0.055 (5.5)	-0.044 (4.6)	-0.031 (4.0)	-0.016 (3.6)

^aNumbers in parentheses are *t*-statistics.

model with random individual-country effects. The Chow (1966) test is usually performed to test the poolability of the data when the error term is homoskedastic. However, since the error term in (10) is not homoskedastic, the Chow test is inappropriate in this case. In fact, Baltagi (1981) showed that if σ_μ^2 is sufficiently large, the Chow test will on the average reject poolability when in fact it is true. The proper test to perform in this case is that described by Roy (1957) and Zellner (1962). This latter test was shown to have a low frequency of type I error, on the average, and was recommended in case pooling by an error component model was contemplated. The Roy-Zellner test applied to (10) rejected the stability of the time-series regression across countries, but not the stability of the cross-country regressions over time.¹² In order to justify pooling, presumably both sets of regressions have to be stable. Yet, as explained above, it was our dissatisfaction with individual-country time-series regressions (see table 1) that prompted us to pool. Like Houthakker (1965), we prefer the results of the pooled model based on *a priori* economic grounds even though formal statistical tests reject poolability. Our view of the importance of such tests is tempered by the fact that any postulated error structure in a pooled model is likely to involve a substantial simplification to the true error structure. Consequently, the resulting power of the test statistics will be poor and the decision based on the application of such poolability tests may be inappropriate. We do not, however, advocate the abandonment of such tests as they are useful indicators of the extent of the underlying heterogeneity of the sample.

Given the decision to pool using an error-component model, we then asked whether the individual-country effects are significant and/or whether the time-period effects are significant. In other words, we asked whether our error-components model is one-way with individual-country effects only, or a two-way model with both individual-country and time-period effects. Under the two-way model, the error term in (10) becomes

$$u_{it} = \mu_i + \lambda_t + v_{it}, \quad (14)$$

where the λ_t 's denote the time-period effects that are independent of each other and the μ_i 's and v_{it} 's. The null hypotheses can now be formulated as follows:

$$H_A: \sigma_\mu^2 = 0, \quad \sigma_\lambda^2 = 0, \quad H_B: \sigma_\mu^2 = 0, \quad H_C: \sigma_\lambda^2 = 0,$$

¹²To illustrate how misleading the conventional Chow test for poolability is, one calculates $F_{85,180} = 99.75$. The appropriate Roy-Zellner test gives $F_{85,180} = 12.22$ using the Wallace and Hussain estimator. Even the latter rejects the null hypothesis, but the extent of rejection is vastly different.

where σ_μ^2 and σ_λ^2 denote the variances of the individual-country effects and the time-period effects, respectively. The proper test to perform in this case is the Lagrange multiplier test described by Breusch and Pagan (1980). This test was applied to (10), and both H_A and H_B were rejected but not H_C . This result supported our *a priori* belief that only country effects are significant and a one-way error-components model rather than a two-way error-components model should be used.

Finally, the Hausman (1978) specification test was employed to test whether the unobservable individual-country effects were uncorrelated with the exogenous variables. The Hausman test is particularly important given the disparate price elasticities indicated for the static model ranging from the within price elasticity of -0.32 vs. the OLS price elasticity of -0.89 . Both the static specification and the sole reliance on within-country variation suggest that the within estimator is likely to reflect essentially a short-run response. In contrast, the OLS estimator depends to a major degree on inter-country variation, which is more likely to reflect long-run adaptations in auto efficiency, utilization, and driving infrastructure.

Comparing European driving conditions with U.S. or Canadian conditions, we observe in Europe much more compact cities, narrow streets, and limited multi-lane highways all of which promote the use of small cars which are used less intensively. These unobservable attributes are of course associated with countries in which gasoline prices are relatively high. Therefore we would expect the Hausman-test to reject the null hypothesis of no correlation between the μ_i 's and the exogenous variables in the static model. In fact, the Hausman chi-squared statistic of 26.9 rejects the null hypothesis. If long-run patterns in auto efficiency, utilization, and driving conditions depend on price effects, we would expect the introduction of a dynamic specification to reduce and possibly eliminate the correlation between these unobservable effects in the error term and the exogenous variables. The Hausman chi-squared statistic for Model III is 9.7, which does not reject the null hypothesis.¹³ This finding appears supportive of the dynamic specification and the value of using inter-country data in estimating long-run responses.

7. Summary and conclusions

With respect to the three methodological questions raised in the introduction, we offer the following conclusions: First, the efficiency gain in this study from pooling is indeed large. For most individual-country time-series models, the coefficient estimates varied widely across countries and

¹³The Hausman test was not applied to Model II (the Koyck-lag model) because of the presence of the lagged dependent variable, which is definitely correlated with the individual-country effects.

exhibited large standard errors. Individual-country time-series models may even lead the researcher to incorrectly conclude the absence of price and/or income effects. In contrast, the pooled sample yields plausible price and income elasticities with low standard errors.

Second, with respect to differences among estimators, all eight pooling estimators tended to yield considerably different coefficients emphasizing the difference between inter and intracountry variation. Of particular interest to applied researchers is the finding that even among the two-stage GLS coefficient estimators, we observe appreciable differences in the coefficients. Even after adopting a dynamic specification to alleviate the tendency of intra-country data to reflect short-run responses, differences in the coefficient estimators persisted, although to a lesser degree.

Finally, our battery of tests on the pooled models emphasize the range of available tests and the importance of applying and interpreting the proper tests. As the results for the poolability tests and the Hausman test for specification error illustrate, the interpretation of these tests in the context of the model postulated raises some interesting and unresolved issues.

With respect to the empirical implications for gasoline demand, the pooled inter-country sample offers considerable support for the view that the long-run price elasticity falls in the range -0.6 to -0.9 . These results emphasize a slow adaptation rate which is of course consistent with Sweeney's view that the major responses to gasoline prices occur through the adaptation of the efficiency characteristics of the auto fleet.

Turning to the broader question of preferred specifications and estimation techniques in the presence of short- and long-run responses, we offer the following recommendations. First, comparison of the within- and between-country estimators in a static specification will reveal whether there exists a short-run or long-run dichotomy. Differences among GLS estimators are likely in this situation and a useful formal test is the Hausman specification test. The researcher should then proceed to a dynamic specification again comparing the OLS, GLS, and within estimators. If the OLS and GLS estimators are similar, the preferred estimator is simply GLS. The researcher faces a more difficult choice when, as in our case, substantial differences still persist between the OLS and GLS estimators, indicating that the country-specific effects are correlated with the explanatory variable. The choice depends critically on the cause for this correlation. If the explanatory variable (price) is slowly but systematically affecting the country-specific effect (driving infrastructure), then OLS may provide better long-run estimators. On the other hand, if this correlation is merely spurious, GLS is the preferred long-run estimator.

Data appendix

As reported by the OECD's *Energy Statistics*, consumption of gasoline

(GAS) is measured as the total consumption of motor gasoline. The motor vehicle variable (CAR), and trucks (TRK) are available from the U.S. *Statistical Yearbook*. Real income per capita (Y/N) is available from the OECD's *National Accounts*. In dealing with international cross-section data there is the problem of translating different currencies to a common denominator. Instead of simply making an exchange-rate conversion, we have used purchasing-power parity-adjustment factors provided by the Pennsylvania International Comparison Project. The purchasing-power parity adjustment affects both real per capita income (Y/N) and the price of other goods (P_{GDP}). The price of motor gasoline (P_{MG}) is, of course, inclusive of taxes, which account for the wide inter-country differences in gasoline prices. Prices of motor gasoline were made available from the U.S. Department of Energy.

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